TASK 3: Protein Structure Prediction

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Title -
Structural Analysis of Human Lysozyme C Using AlphaFold and PyMOL

Selected protein: Lysozyme C

Gene: LYZ

Organism: Homo sapiens (Human)

UniProt accession: P61626

Sequence length: 148 amino acids

Predicted structure: AlphaFold Protein Structure Database

Visualization software: PyMOL

Aim
To obtain the amino-acid sequence of a selected protein from UniProt, use an AlphaFold-predicted three-dimensional structure, visualize the model using PyMOL, and describe its major structural features.

Protein Selection

Human Lysozyme C was selected because it is a small, well-characterized globular enzyme with a clearly defined fold and biologically important active-site residues. The UniProtKB reviewed entry P61626 identifies the protein as Lysozyme C from Homo sapiens and reports a canonical sequence of 148 amino acids.

UniProt Sequence

UniProt accession: P61626

FASTA sequence:

MKALIVLGLVLLSVTVQGKVFERCELARTLKRLGMDGYRGISLANWMCLAKWESGYNTRA
TNYNAGDRSTDYGIFQINSRYWCNDGKTPGAVNACHLSCSALLQDNIADAVACAKRVVRD
PQGIRAWVAWRNRCQNRDVRQYVQGCGV

Structure Prediction Method

The three-dimensional structure was obtained from the AlphaFold Protein Structure Database using the UniProt accession P61626. AlphaFold predicts protein structure from the amino-acid sequence. The downloaded structure file is then loaded into PyMOL for visualization and preparation of publication-style structure images.

The structure should be treated as a computational prediction rather than a newly determined experimental structure. AlphaFold confidence information (such as pLDDT) should be checked on the AlphaFold entry when discussing model reliability.

Structural Features

- **Overall fold:** Human Lysozyme C is a compact globular protein composed of two structural regions/domains: an α-helix-rich α-domain and a smaller β-domain.
- **Mature protein:** The first 18 residues form a signal peptide. The mature secreted protein corresponds to residues 19-148 and contains 130 amino acids.
- **α-helices:** The α-domain contains four principal α-helices. These helices contribute strongly to the compact globular fold.
- **β-structure:** The β-domain contains a three-stranded antiparallel β-sheet together with connecting loops.
- **Disulfide bonds:** The mature protein contains four disulfide bonds that help stabilize the folded structure. In full UniProt numbering the cysteine pairs are Cys24-Cys146, Cys48-Cys134, Cys83-Cys99 and Cys95-Cys113.
- **Active-site cleft:** A prominent cleft lies between the α- and β-regions and accommodates polysaccharide substrate.
- **Catalytic residues:** The key catalytic residues are Glu35 and Asp52 in mature-protein numbering. Because UniProt P61626 includes an 18-residue signal peptide, these correspond to Glu53 and Asp70 in full-length UniProt numbering.
- **Function:** Lysozyme C hydrolyzes β(1→4) linkages in bacterial peptidoglycan, contributing to antibacterial defense.

PyMOL Visualization

The supplied PyMOL script produces three recommended views: an overall cartoon representation, an active-site view with catalytic residues shown as sticks, and a surface representation showing the active-site region.

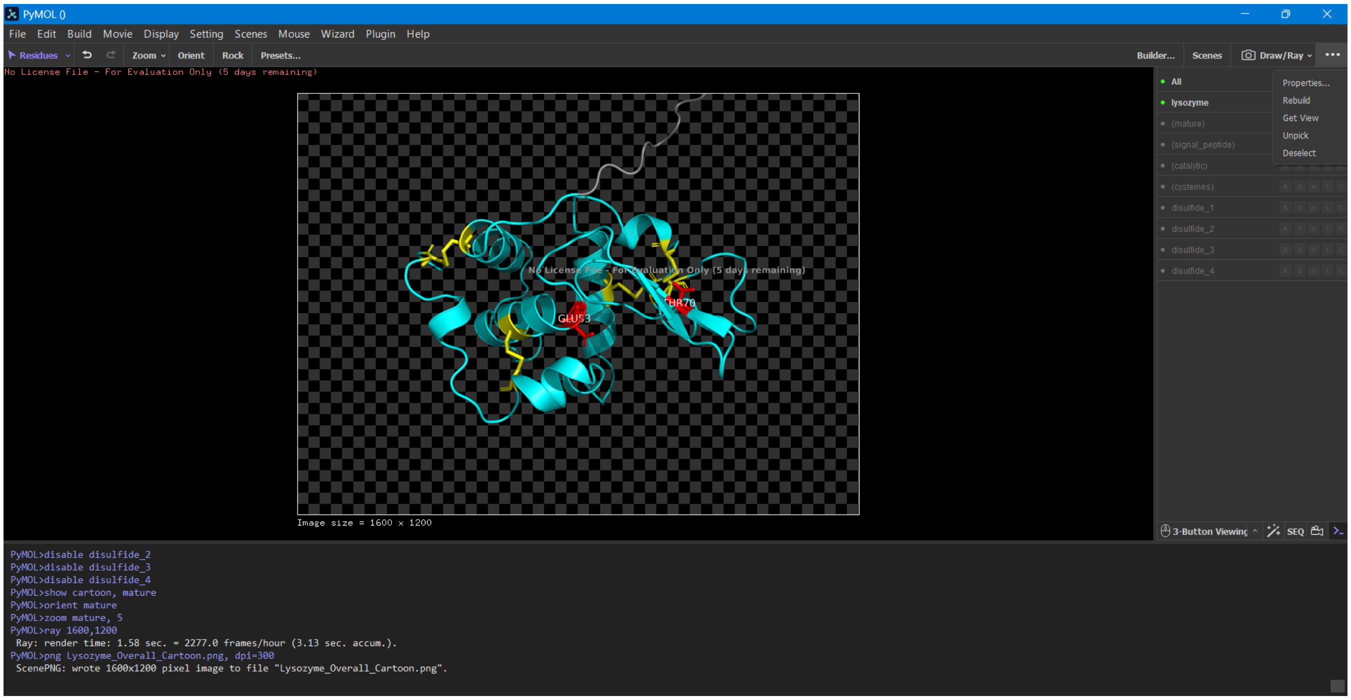

FIGURE 1

Overall cartoon representation of the AlphaFold-predicted structure of human Lysozyme C.

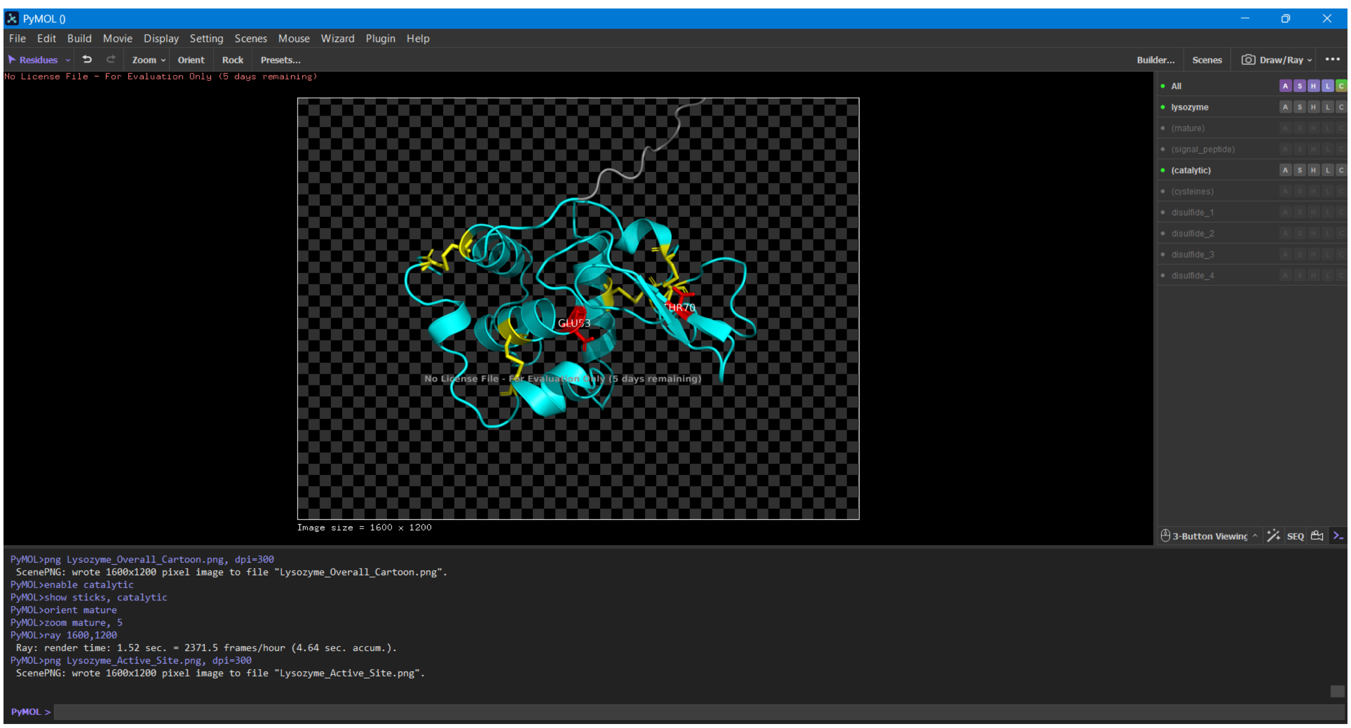

FIGURE 2

PyMOL visualization of human Lysozyme C showing the predicted fold and the catalytic residues in the active-site region.

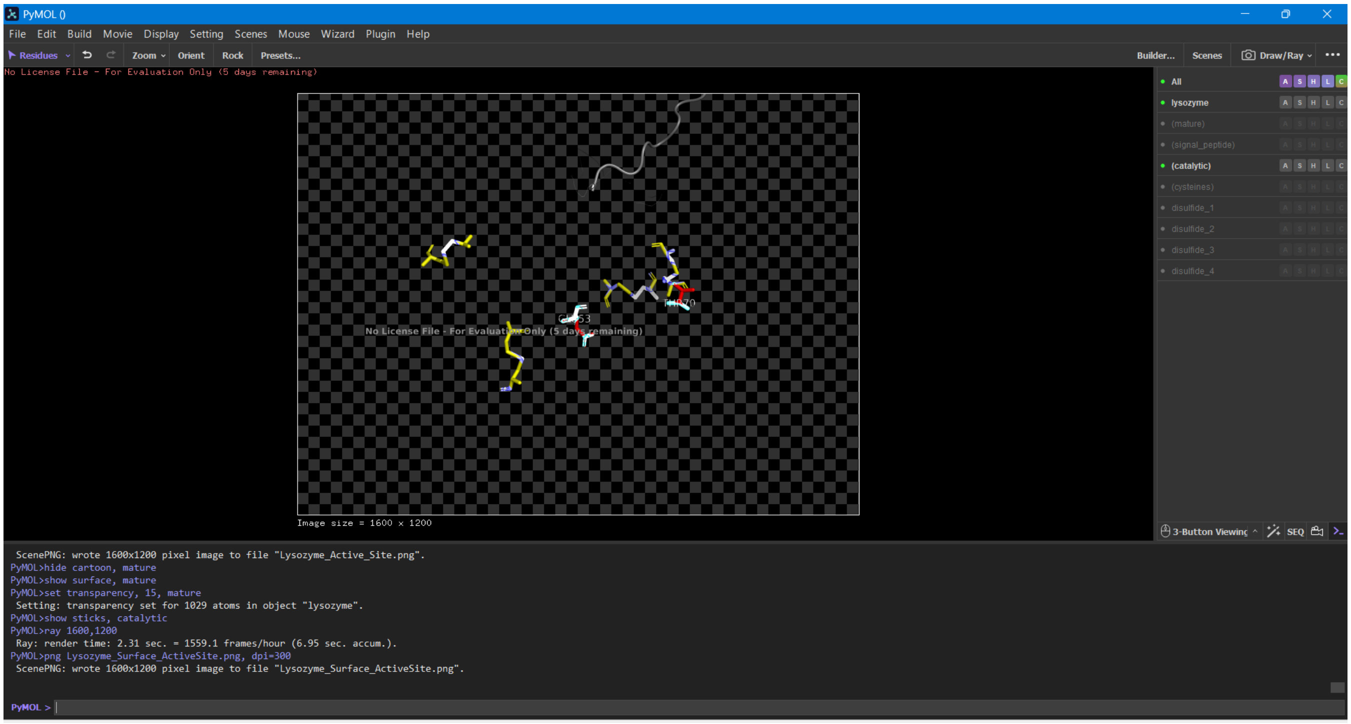

FIGURE 3

Surface representation of human Lysozyme C illustrating the active-site cleft and location of the catalytic residues.

Interpretation

The predicted structure shows the characteristic compact lysozyme fold. The α-helical and β-sheet regions are arranged to form a deep cleft. This cleft is functionally important because it provides the binding environment for polysaccharide substrates. The catalytic acidic residues are positioned within this region and are therefore structurally linked to enzymatic activity.

The four disulfide bonds provide additional stabilization, particularly for the loops and inter-domain arrangement of the mature protein. The combination of α-helices, β-sheet, loops, and disulfide bonds produces a stable globular structure suitable for its extracellular antibacterial role.

Result

A three-dimensional structural model of human Lysozyme C (UniProt P61626) was obtained from the AlphaFold Protein Structure Database and visualized using PyMOL. The model displays the expected globular architecture with an α-helix-rich region, a β-sheet-containing region, a substrate-binding cleft, four disulfide bonds, and a conserved catalytic site.

Conclusion

Protein structure prediction provides a useful way to relate amino-acid sequence to three-dimensional architecture and biological function. For human Lysozyme C, the AlphaFold model allows the major secondary and tertiary structural features to be visualized and connected to its enzymatic antibacterial function. PyMOL further allows important residues and structural features to be highlighted for interpretation.

Appendix A – Complete PyMOL Code –

```
# 1. Download the AlphaFold PDB for P61626.
# 2. Save it in the same folder as this script as:
#      AF-P61626-F1-model_v4.pdb
# 3. Open this .pml file in PyMOL.

reinitialize

load AF-P61626-F1-model_v4.pdb, lysozyme

# Basic appearance
hide everything, lysozyme
show cartoon, lysozyme
set cartoon_fancy_helices, 1
set cartoon_sampling, 14

# Focus on the mature protein (UniProt residues 19-148)
select mature, lysozyme and resi 19-148
show cartoon, mature
color cyan, mature

# Signal peptide: residues 1-18
select signal_peptide, lysozyme and resi 1-18
show cartoon, signal_peptide
color grey, signal_peptide

# Catalytic residues in full-length UniProt numbering:
# Mature Glu35 = UniProt Glu53
# Mature Asp52 = UniProt Asp70
select catalytic, lysozyme and resi 53+70
show sticks, catalytic
color red, catalytic
label catalytic and name CA, "%s%s" % (resn,resi)
set label_size, 16

# Four disulfide bonds (full-length numbering)
select cysteines, lysozyme and resi 24+48+83+95+99+113+134+146
show sticks, cysteines
color yellow, cysteines

# Optional: show disulfide-connected cysteine pairs as dashed bonds
distance disulfide_1, lysozyme and resi 24 and name SG, lysozyme and resi 146 and name SG
```

```
distance disulfide_2, lysozyme and resi 48 and name SG, lysozyme and resi 134 and name SG
distance disulfide_3, lysozyme and resi 83 and name SG, lysozyme and resi 99 and name SG
distance disulfide_4, lysozyme and resi 95 and name SG, lysozyme and resi 113 and name SG
set dash_color, yellow
set dash_width, 3

# Active-site cleft view
orient mature
zoom mature, 5

# ---------- IMAGE 1: Overall cartoon ----------
disable catalytic
disable cysteines
disable disulfide_1
disable disulfide_2
disable disulfide_3
disable disulfide_4
show cartoon, mature
ray 1600, 1200
png Lysozyme_Overall_Cartoon.png, dpi=300

# ---------- IMAGE 2: Cartoon + catalytic residues ----------
enable catalytic
show sticks, catalytic
ray 1600, 1200
png Lysozyme_Active_Site.png, dpi=300

# ---------- IMAGE 3: Surface + catalytic residues ----------
hide cartoon, mature
show surface, mature
set transparency, 15, mature
show sticks, catalytic
ray 1600, 1200
png Lysozyme_Surface_ActiveSite.png, dpi=300

# Return to a clean presentation view
hide surface, mature
show cartoon, mature
enable cysteines
enable disulfide_1
enable disulfide_2
enable disulfide_3
enable disulfide_4
orient mature
zoom mature, 5

# Save a PyMOL session
save Task3_Lysozyme_P61626.pse
```
